
Global Convergence and High-Frequency Spectral Bias of Low-Rank Networks

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Low-rank neural networks are usually introduced as compressed versions of dense
2 models, but this view misses their main theoretical role: rank creates a structured
3 mean-field system whose optimization and spectral behavior can be analyzed
4 directly. We prove that low-rank random-feature neural networks, with frozen
5 feature maps and trainable channel weights, have well-posed mean-field dynamics
6 and that every convergent trajectory reaches a global minimizer of the population
7 risk under a standard loss-identifiability condition. The result holds for arbitrary
8 depth and standard independent initialization; low rank changes constants through
9 the channel mixing norm but does not destroy the dense-span argument needed
10 for global optimality. We then explain why convergence alone is not enough for
11 highly oscillatory functions. Building on the spectral-bias perspective of Rahaman
12 et al. [17], low rank is shown to control path-space degrees of freedom, continuous
13 piecewise affine geometry, and finite-time Fourier recovery. Experiments on
14 oscillatory targets and Fourier-native rank sweeps support the view that the useful
15 rank is often intermediate: small enough to change spectral transfer, but large
16 enough to avoid an approximation bottleneck.

17 1 Introduction

18 Low-rank neural networks replace dense matrices by factored or bottlenecked operators. The standard
19 motivation is computational: fewer trainable parameters and cheaper memory traffic. This paper
20 argues that the more important question is mathematical. Does the low-rank constraint preserve the
21 optimization guarantees available in mean-field theory, and can the same constraint help with the
22 spectral bias that prevents neural networks from learning highly oscillatory functions?

23 We answer the first question positively for a low-rank random-feature architecture. The model
24 freezes a rich random-feature map and trains low-rank channel weights through mean-field gradient
25 flow. Since the frozen features keep a dense span throughout training, the usual obstruction in deep
26 mean-field convergence proofs is removed: at a limit point, zero gradient implies the conditional
27 first-order condition for the loss, and the loss-identifiability assumption forces global optimality. Low
28 rank enters through the mixing matrix and channel index, changing constants but not the logic of the
29 proof.

30 The second question is more subtle. A globally convergent training dynamic can still learn low
31 frequencies first and fail at finite time on highly oscillatory targets. We therefore connect the
32 convergence theorem to a rank-dependent spectral bias mechanism. For ReLU networks, low
33 rank constrains active path coefficient tensors; this constrains switching normals, suppresses high-
34 codimension faces in the continuous piecewise affine complex, and changes Fourier shell behavior.
35 In one dimension, the raw spline tail remains close to the classical k^{-4} law, so the correct diagnostic
36 is not the asymptotic exponent alone but the learned Fourier recovery curve. This leads to a practical

rank-selection rule: choose the smallest rank that avoids the approximation bottleneck while flattening the finite-time spectral transfer.

Contributions. The paper makes three contributions:

- We state a global convergence theorem for arbitrary-depth low-rank random-feature networks in the mean-field limit.
- We turn low-rank spectral bias into a geometric statement: rank constrains active path coefficients, which constrains CPWL switching geometry and Fourier shell energy.
- We connect this mechanism to experiments on highly oscillatory targets, including high-frequency regression, frozen-vs-trainable random features, and heavy-tail spectral updates.

2 Related Work

Mean-field convergence. Mean-field theory analyzes neural network training through the deterministic evolution of parameter distributions in the large-width limit. Foundational work established global convergence mechanisms for two-layer models and Wasserstein gradient flows under suitable convexity, homogeneity, or support assumptions [13, 5]. Later multilayer mean-field analyses clarified how deep feature maps evolve and when stationary points can be related to global optima [16, 15]. Our result follows this line but focuses on low-rank random-feature networks: the existence and uniqueness estimates use the same mean-field template, while the convergence proof must be adapted to channel-wise low-rank couplings.

Low-rank and random features. Low-rank neural networks are often motivated by parameter efficiency, with LoRA-style decompositions now standard in large-scale adaptation [12]. Most theory for low-rank networks focuses on expressivity, approximation, or kernel behavior rather than global training dynamics. Random features provide a tractable bridge between finite networks and kernel methods: fixing the feature map turns the representation into a stable basis, while training the channel weights retains a mean-field optimization problem. In our RF-LR setting, frozen random features prevent support collapse and preserve the dense-span property needed for the stationarity-to-optimality argument; the low-rank factors enter through bounded channel mixing rather than through a full dense contraction chain.

Bottleneck rank and implicit low-rank bias. Recent work gives a functional notion of optimal rank that is closer to our setting than the algebraic rank of a single matrix. Jacot [9] studies the implicit bias of large-depth homogeneous networks and shows that their representation cost converges toward a nonlinear rank notion, sandwiched between Jacobian rank and bottleneck rank. The bottleneck rank is the smallest inner dimension k for which a function can be factored as $f = h \circ g$ through $\mathbb{R}^k$. Follow-up work [10] proves that learned deep representations exhibit a bottleneck structure and introduces finite-depth corrections that balance low inner dimension against regularity. This suggests a theoretical target rank: the smallest rank that captures the intrinsic bottleneck dimension of the target without forcing an irregular or rank-underestimating interpolant. Complementarily, Bantzis, Simon, and Jacot [3] show that the first saddle escape of deep ReLU networks has a low-rank bias in deeper layers and propose saddle-to-saddle dynamics with increasing bottleneck rank. Our rank-selection principle is consistent with this picture: training should start from low effective rank, but useful learning requires increasing rank until approximation and spectral recovery saturate.

Spectral bias. The spectral-bias phenomenon was formalized by Rahaman et al. [17], who showed that neural networks tend to learn low-frequency components before high-frequency components and related this behavior to Fourier decay properties of piecewise linear networks. This paper uses that viewpoint but changes the architectural knob: instead of only observing that high frequencies are learned late, we ask how rank changes the CPWL geometry that controls the finite-resolution Fourier transfer. The key difference is coefficient-level: low rank suppresses high-codimension face moments in the Fourier shell expansion, moving the observed spectrum toward the facet-dominated regime discussed by Rahaman et al. [17].

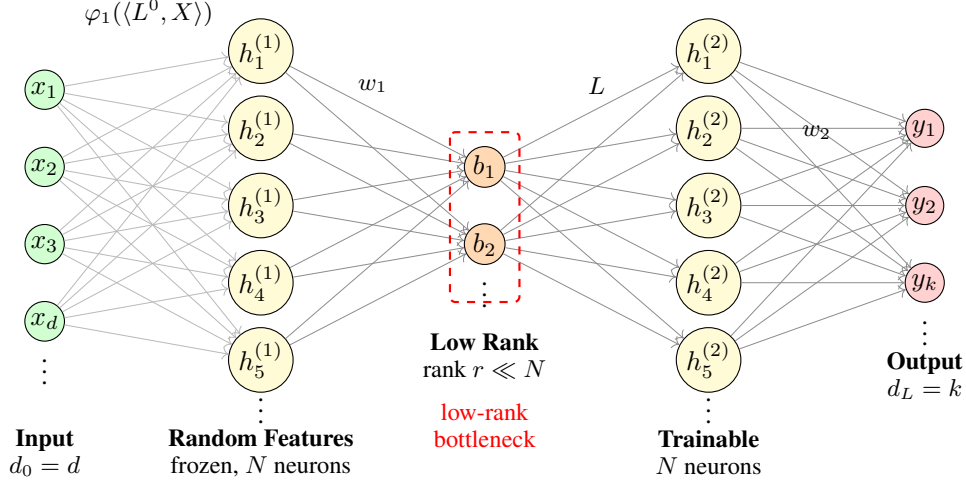


Figure 1: Architecture of a three-layer low-rank random-feature network. The first layer consists of frozen random features, the second layer employs a low-rank mixing matrix L of rank $r \ll N$, and the third layer contains trainable weights w_2 . The red dashed box highlights the low-rank bottleneck that reduces dimensionality from N to r channels.

3 Freezing Some Weights Allows Global Convergence

Let $X \in \mathbb{R}^d$ and Y be drawn from a population distribution. The finite-width low-rank random-feature network used throughout the paper is the one from the original experiments:

$$h^{(0)}(x) = x, \quad h^{(\ell)}(x) = \frac{1}{n_\ell} \sum_{j=1}^{n_\ell} w_j^{(\ell)} \varphi_\ell \left(\langle L_j^{(\ell)}, h^{(\ell-1)}(x) \rangle + b_j^{(\ell)} \right), \quad (1)$$

where the feature directions $L_j^{(\ell)}$ and biases $b_j^{(\ell)}$ are frozen, while the channel weights $w_j^{(\ell)}$ are trained. This is a low-rank factorization of a dense layer in which the left factor is fixed: if $W_\ell = U_\ell V_\ell^\top$ with $U_\ell \in \mathbb{R}^{n_\ell \times r_\ell}$ bounded and full column rank, then U_ℓ plays the role of a frozen random-feature or mixing basis and the trainable factor V_ℓ is absorbed into the channel weights. Thus one may avoid training U_ℓ entirely; drawing it as a bounded full-rank tall-skinny matrix is enough for the theory, while choosing $U_\ell \in \text{St}(n_\ell, r_\ell)$ is an optional numerical normalization.

3.1 Mean-Field Formulation

The mean-field viewpoint is the relevant one for global convergence. Pure kernel or neural-tangent analyses give convexity by freezing the representation, but they largely remove feature learning. Finite-width nonconvex analyses allow feature learning but rarely provide global convergence guarantees for deep low-rank networks. Mean-field theory is currently the most developed framework that can keep a feature-learning interpretation while still proving that limiting gradient-flow trajectories reach global minimizers under identifiability assumptions. We use a frozen-feature low-rank specialization because it is the cleanest setting in which the dense-span argument is preserved; it should be viewed as the globally convergent anchor case for more general low-rank feature-learning models.

In the mean-field limit, frozen feature maps are written $L^0(C_\ell)$ and trainable channel weights are written $w_\ell(t, C_\ell)$. The channel mixer $L \in \mathbb{R}^{N_2 \times r}$ is a bounded full-column-rank tall-skinny factor, not necessarily an orthogonal matrix. We write the second-layer operator as $W_2 = L A_2^\top$, with L fixed and A_2 trainable, then absorb the small right factor $A_2^\top$ into the rank-channel functions below. One may additionally choose $L \in \text{St}(N_2, r)$, or periodically re-orthogonalize it, as a numerical design choice; this is natural for matrix-structured optimizers such as Muon, which orthogonalize matrix updates by Newton–Schulz or polar iterations [11, 4]. In the two-hidden-layer case, the first partial functions are

$$f_k(t, x) = \mathbb{E}_{C_1} [w_1(t, C_1, k) \varphi_1(\langle L^0(C_1), x \rangle)], \quad k = 1, \dots, r, \quad (2)$$

111 and the second-layer preactivation is

$$H_2(t, c_2; x) = \sum_{k=1}^r L_{c_2, k} f_k(t, x). \quad (3)$$

112 Thus H_2 is the reconstruction of the second-layer signal from a bounded rank- r channel basis. The
 113 output is obtained by averaging the trainable top weights against $\varphi_2(H_2)$. For deeper networks,
 114 the same alternating pattern is used: trainable channel weights are averaged against frozen random
 115 features and mixed through bounded low-rank channel matrices.

116 The population objective is

$$\mathcal{L}(W) = \mathbb{E}_{(X, Y)} [\mathcal{L}(Y, \hat{y}(X; W))]. \quad (4)$$

117 The mean-field gradient flow evolves the trainable weights by

$$\partial_t w_\ell(t, \cdot) = -\xi_\ell(t) \nabla_{w_\ell} \mathcal{L}(W(t)), \quad (5)$$

118 where ξ_ℓ are bounded learning-rate schedules. The exact formulas are standard backpropagation in
 119 the mean-field variables; the only low-rank modification is the channel mixing through $L_{c_\ell, k}$.

120 Under the technical regularity, diversity, loss-identifiability, and convergence assumptions stated in
 121 Appendix A, freezing the feature directions gives a well-defined mean-field system and allows the
 122 usual stationarity argument to conclude global optimality.

123 **Theorem 3.1** (Well-posedness and global convergence of RF-LR training). *Under Assumptions A.1–*
 124 *A.6, the low-rank mean-field gradient-flow system has a unique solution on every finite interval $[0, T]$*
 125 *and therefore on $[0, \infty)$. Moreover, if the resulting low-rank mean-field dynamics converges to W^* ,*
 126 *then W^* is a global minimizer of the population loss:*

$$\mathcal{L}(W^*) = \inf_W \mathcal{L}(W). \quad (6)$$

127 For every depth $L \geq 2$, this holds with standard independent initialization of the trainable weights.

128 The well-posedness part only justifies that the limiting dynamics are genuine ODEs rather than formal
 129 flows. The technical fixed-point estimates are standard mean-field estimates with the dense-layer
 130 constants replaced by low-rank mixing norms, and are not the focus of the paper.

131 *Proof idea.* The existence and uniqueness part follows the standard mean-field fixed-point argument.
 132 We only summarize the convergence-to-global-minimum step here; Appendix B gives the full notation
 133 and the five formal steps. We assume that gradient descent, or its mean-field gradient-flow limit, has
 134 converged somewhere: $W(t) \rightarrow \bar{W}$ in the coupling topology. Stationarity and the persistent dense
 135 span of the frozen first-layer features then imply the integrated identity

$$\mathbb{E}[d_L(Z; \bar{W}) \mid X = x] B_k^{(2)}(x; \bar{W}) = 0 \quad \text{for } \mathbb{P}_X\text{-a.e. } x, \quad k = 1, \dots, r, \quad (7)$$

136 where $d_L(Z; \bar{W}) = \partial_2 \mathcal{L}(Y, \hat{y}(X; \bar{W}))$ and $B_k^{(2)}$ is the channel-wise backpropagated factor defined
 137 in Appendix B.

138 *Step 3: remove the backpropagated factor.* The non-degeneracy assumptions exclude a dead-channel
 139 limit: for $\mathbb{P}_X$ -a.e. x , at least one channel has a nonzero backpropagated factor $B_k^{(2)}(x; \bar{W})$, and
 140 the analogous statement holds layer by layer for $L > 3$. For ReLU-type activations, the forward
 141 estimates only need the channel Lipschitz inequality

$$\left| \varphi\left(\sum_{k=1}^r x_k\right) - \varphi\left(\sum_{k=1}^r y_k\right) \right| \leq \sum_{k=1}^r |x_k - y_k|. \quad (8)$$

142 Consequently (7) implies

$$\mathbb{E}[\partial_2 \mathcal{L}(Y, \hat{y}(X; \bar{W})) \mid X = x] = 0 \quad \text{for } \mathbb{P}_X\text{-a.e. } x. \quad (9)$$

143 *Step 5: the trajectory loss converges to the limit loss.* Since $W(t) \rightarrow \bar{W}$ in the coupling topology,
 144 there are couplings π_t between $W(t)$ and $\bar{W}$ whose weighted channel gaps vanish. In the two-layer
 145 channel notation, the key convergence statement is exactly

$$\mathbb{E}_{\pi_t} \left[(1 + |\bar{w}_2|) |\bar{w}_2| \sum_{k=1}^r |\bar{w}_{1, k}| |w_{1, k} - \bar{w}_{1, k}| \right] \rightarrow 0, \quad (10)$$

with analogous quantities for all deeper layers. The low-rank form $H_2 = \sum_k L_{c_2,k} f_k$ bounds every forward and backward difference by these same channel integrals. Hence

$$\mathbb{E} [|\hat{y}(X; W(t)) - \hat{y}(X; \bar{W})|] \rightarrow 0. \quad (11)$$

Lipschitzness of the loss in its second argument on the a priori bounded trajectory yields

$$|\mathcal{L}(W(t)) - \mathcal{L}(\bar{W})| \leq K \mathbb{E} [|\hat{y}(X; W(t)) - \hat{y}(X; \bar{W})|] \rightarrow 0. \quad (12)$$

Step 4: identify the limit as globally optimal. By the loss-identifiability condition,

$$\mathbb{E}[\partial_2 \mathcal{L}(Y, u) \mid X = x] = 0 \implies \mathbb{E}[\mathcal{L}(Y, u) \mid X = x] = 0. \quad (13)$$

Applying this with $u = \hat{y}(x; \bar{W})$ and using non-negativity of $\mathcal{L}$ gives

$$\mathcal{L}(\bar{W}) = 0 = \inf_W \mathcal{L}(W). \quad (14)$$

Together with Step 5, $\mathcal{L}(W(t)) \rightarrow \mathcal{L}(\bar{W})$. $\square$

4 Low-Rank Spectral Bias

Related work: spectral bias. Theorem 3.1 says that any convergent trajectory reaches a global minimizer. It does not say how fast different frequencies are learned, nor which finite-time solution is reached under a fixed training budget. Highly oscillatory functions expose this gap. This is precisely the phenomenon called spectral bias by Rahaman et al. [17]: standard neural networks tend to fit low-frequency components before high-frequency components. Rahaman et al. [17] explain this behavior through Fourier decay and training dynamics of neural networks.

Our mechanism is complementary and more structural. In ReLU networks, the represented function is continuous piecewise affine (CPWL), so Fourier behavior can be related to the polyhedral complex induced by activation boundaries. Low rank does not merely reduce parameter count; it constrains the algebraic degrees of freedom of the active path coefficients, and this changes which CPWL faces can carry Fourier mass. Our goal is therefore not to contradict classical spectral bias, but to identify rank as an architectural knob that modifies the geometry behind the finite-time Fourier transfer.

In a ReLU network, on every activation region the network is affine, and the affine coefficient is a sum over active paths. The high-level idea is that a low-rank network has fewer *independent* paths in this sum-over-paths representation. More precisely, when $W_\ell = U_\ell V_\ell^\top$ with rank r_ℓ , the path coefficients involving layer ℓ must factor through an r_ℓ -dimensional latent coordinate. Thus the graph may still contain many symbolic paths, but the number of independent path constraints is much smaller than in a full-width layer. Fewer independent path constraints mean fewer independent switching constraints for the polytopes of the CPWL linear-region complex. Geometrically, this favors more high-dimensional faces and suppresses low-dimensional, high-codimension intersections. Spectrally, the Fourier shell law below then predicts a lower finite-resolution decay, closer to the facet-dominated regime.

Mechanism summary. The chain of implications is the following. A factorization $W_\ell = U_\ell V_\ell^\top$ does not only compress parameters; it forces all layer- ℓ directions to pass through an r_ℓ -dimensional bottleneck. This reduces the number of independent path coefficients, which reduces the number of independent switching normals that can cut the input space. The CPWL complex then contains fewer stable high-codimension intersections and relatively more facet-dominated contributions. Since the Fourier expansion of a CPWL function is organized by codimension, suppressing high-codimension face moments moves the finite-resolution spectrum toward the slow-decay facet regime described by Rahaman et al. [17]. The useful rank is therefore not the smallest rank: if r is too small, approximation error dominates; if r is too large, the network recovers the full-rank spectral bias. The target is the first intermediate rank where approximation is controlled and the Fourier recovery scaling has already saturated.

This gives a theoretical way to define an optimal rank. At the population level, the natural lower bound is the bottleneck rank of the target, $\text{Rank}_{BN}(f^*)$ [9, 10]. Under finite training time and finite data, the practical rank should minimize a three-term trade-off

$$r^*(T, \varepsilon) \in \arg \min_r \{ \mathcal{A}_r(f^*) + \lambda_T \mathcal{O}_r(T) + \lambda_F \mathcal{S}_r \}, \quad (15)$$

where $\mathcal{A}_r(f^*)$ is the rank- r approximation error, $\mathcal{O}_r(T)$ is an optimization cost at time T , and $\mathcal{S}_r$ is the spectral-bias cost measured by Fourier recovery or CPWL face moments. In experiments, this criterion is estimated by the first rank for which approximation error is small and the log-log Fourier recovery slope no longer improves. Thus the predicted optimum is a saturation rank, not full rank.

This algebraic constraint becomes geometric. Inside a cell of the previous layers, a new switching normal has the form

$$n_{\ell i \varepsilon} = B_{\ell-1, \varepsilon}^\top V_\ell u_{\ell i} \in \text{Im}(B_{\ell-1, \varepsilon}^\top V_\ell), \quad \dim \text{Im}(B_{\ell-1, \varepsilon}^\top V_\ell) \leq \min(d, r_1, \dots, r_\ell). \quad (16)$$

Thus rank lowers the dimension of the normal subspace and suppresses high-codimension faces first.

Face Fourier expansion. Let $g = \chi f$, where f is CPWL on a finite polyhedral complex in $\mathbb{R}^d$ and $\chi \in C_c^\infty(\mathbb{R}^d)$ is an observation window. Let $\mathcal{F}_q(f)$ be the codimension- q faces of the complex. For every large frequency $\xi = \rho\theta$, $\theta \in \mathbb{S}^{d-1}$, the Fourier transform has the expansion

$$\widehat{g}(\rho\theta) = \sum_{q=1}^d \rho^{-(q+1)} \sum_{F \in \mathcal{F}_q(f)} A_F(\rho, \theta) + O(\rho^{-(d+2)}), \quad (17)$$

where A_F is a bounded oscillatory face coefficient controlled by the jump Δ_F of the appropriate normal derivative across the local fan around F . In particular,

$$|A_F(\rho, \theta)| \leq C_\chi \|\Delta_F\| \text{vol}_{d-q}(F) \omega_F(\theta). \quad (18)$$

This is the anisotropic phenomenon isolated by Rahaman et al. [17]: in almost all directions a ReLU network has fast $k^{-(d+1)}$ amplitude decay, while in special directions orthogonal to facets of the linear regions the decay can be as slow as k^{-2} . Thus low codimension faces are precisely the geometric structures that keep high-frequency Fourier mass visible at finite resolution.

Proposition 4.1 (Fourier shell law for CPWL networks). *Let $g = \chi f$ be a windowed continuous piecewise affine network in dimension d . Define the codimension- q face moment*

$$M_q(f) = \sum_{F \in \mathcal{F}_q(f)} \|\Delta_F\|^2 \text{vol}_{d-q}(F)^2 \mathbb{E}_{\theta \in \mathbb{S}^{d-1}} \omega_F(\theta)^2. \quad (19)$$

Then the shell-averaged Fourier energy satisfies

$$S_f(\rho) = \mathbb{E}_{\|\xi\| \simeq \rho} |\widehat{g}(\xi)|^2 \lesssim \sum_{q=1}^d M_q(f) \rho^{-2(q+1)}. \quad (20)$$

Low rank suppresses M_q for larger q first, moving finite-resolution spectra toward the facet-dominated ρ^{-4} energy regime. This should be read as a coefficient statement rather than a claim that low rank creates a new universal exponent: the useful comparison with full rank is the reduction of M_q/M_1 and of the transition coefficient $(M_q/M_1)\rho^{-2q+2}$ for $q \geq 2$.

Proof. Write the CPWL network on its polyhedral complex as

$$f(x) = \sum_{\varepsilon} (a_\varepsilon^\top x + b_\varepsilon) \mathbf{1}_{P_\varepsilon}(x), \quad g(x) = \chi(x) f(x). \quad (21)$$

The cutoff makes g compactly supported, so

$$\widehat{g}(\rho\theta) = \sum_{\varepsilon} \int_{P_\varepsilon} e^{-i\rho\langle\theta, x\rangle} \chi(x) (a_\varepsilon^\top x + b_\varepsilon) dx \quad (22)$$

is an ordinary oscillatory integral. Set $D_\theta = \theta \cdot \nabla$. Since $D_\theta e^{-i\rho\langle\theta, x\rangle} = -i\rho e^{-i\rho\langle\theta, x\rangle}$, the divergence theorem on each cell gives

$$\begin{aligned} \int_{P_\varepsilon} e^{-i\rho\langle\theta, x\rangle} u_\varepsilon(x) dx &= \frac{i}{\rho} \int_{\partial P_\varepsilon} e^{-i\rho\langle\theta, x\rangle} u_\varepsilon(x) \langle\theta, n_\varepsilon(x)\rangle d\sigma(x) \\ &\quad - \frac{i}{\rho} \int_{P_\varepsilon} e^{-i\rho\langle\theta, x\rangle} D_\theta u_\varepsilon(x) dx, \end{aligned} \quad (23)$$

where $u_\varepsilon(x) = \chi(x)(a_\varepsilon^\top x + b_\varepsilon)$ and n_ε is the outward normal. When the boundary terms are summed over all cells, every interior facet $F = P_\varepsilon \cap P_{\varepsilon'}$ is counted twice with opposite normals. The order- ρ^{-1} trace contribution is therefore proportional to

$$(u_\varepsilon|_F - u_{\varepsilon'}|_F) \langle \theta, n_F \rangle. \quad (24)$$

Because f is continuous and the same smooth window χ multiplies both traces, this term vanishes on internal facets. Boundary terms on the exterior of the observation window are smooth cutoff terms and can be integrated by parts arbitrarily many times, hence are absorbed in the final remainder.

After this cancellation, the leading non-smooth contribution comes from applying the same identity to the interior integral involving $D_\theta u_\varepsilon$. On a facet F shared by two cells, its jump is

$$[D_\theta u]_F = \chi \theta^\top (a_\varepsilon - a_{\varepsilon'}) \quad (25)$$

because the terms involving derivatives of χ multiply the continuous trace of f and cancel across F . Thus the first non-canceling boundary term is a jump of directional derivatives, and it carries the factor ρ^{-2} . This gives the codimension-one part of (17). In the special directions where θ is almost normal to a facet, this term is the slow $|\xi|^{-2}$ amplitude contribution emphasized by Rahaman et al. [17].

The higher-codimension terms follow by induction on the face lattice. Assume that after m reductions the surviving terms are sums over codimension- m faces E of the form

$$\rho^{-(m+1)} \int_E e^{-i\rho\langle\theta,x\rangle} B_E^{(m)}(\theta, x) d\sigma_E(x), \quad (26)$$

where $B_E^{(m)}$ is a linear combination of jumps of m -th normal derivatives in the local fan around E , multiplied by derivatives of χ of bounded order. If E is not yet a zero-dimensional face, integrate by parts tangentially on each cell of the induced polyhedral decomposition of E . Tangential interior terms either gain another power of ρ^{-1} and remain smooth, or cancel between adjacent induced cells. The non-canceling terms live on the boundary of E , hence on faces of codimension $m+1$, and their coefficients are precisely the next normal derivative jumps around the refined local fan. Therefore, after q boundary reductions plus the initial affine derivative, every codimension- q contribution has size $\rho^{-(q+1)}$ and is localized on faces $F \in \mathcal{F}_q(f)$.

Consequently the remaining integral over a codimension- q face has the form

$$A_F(\rho, \theta) = \int_F e^{-i\rho\langle\theta,x\rangle} B_F(\theta, x) d\sigma_F(x), \quad (27)$$

where B_F is linear in the jump Δ_F of the appropriate normal derivative and in derivatives of the window χ . Hence

$$|A_F(\rho, \theta)| \leq C_\chi \|\Delta_F\| \text{vol}_{d-q}(F) \omega_F(\theta). \quad (28)$$

The induction stops after codimension d . Terms in which all differentiations fall on the smooth cutoff, or smooth tangential remainders after the last reduction, are $O(\rho^{-(d+2)})$ by one additional integration by parts. This proves the face expansion (17). It also recovers the same anisotropic mechanism as the spectral-decay formula of Rahaman et al. [17]: generic directions have fast $k^{-(d+1)}$ amplitude decay, but directions orthogonal to facets can decay as slowly as k^{-2} .

Squaring the face expansion gives diagonal face terms and oscillatory cross terms. The diagonal contribution of codimension- q faces is bounded by

$$\rho^{-2(q+1)} \sum_{F \in \mathcal{F}_q(f)} \|\Delta_F\|^2 \text{vol}_{d-q}(F)^2 \omega_F(\theta)^2. \quad (29)$$

The off-diagonal terms are oscillatory products between distinct faces. After averaging over a shell, pairs whose phases have no stationary angular alignment are lower order by non-stationary phase. The remaining aligned pairs are controlled by Cauchy–Schwarz:

$$2|A_F A_{F'}| \leq |A_F|^2 + |A_{F'}|^2, \quad (30)$$

so they are absorbed into the same face moments up to a dimension- and complexity-dependent constant. Taking the angular expectation yields

$$S_f(\rho) \lesssim \sum_{q=1}^d M_q(f) \rho^{-2(q+1)}. \quad (31)$$

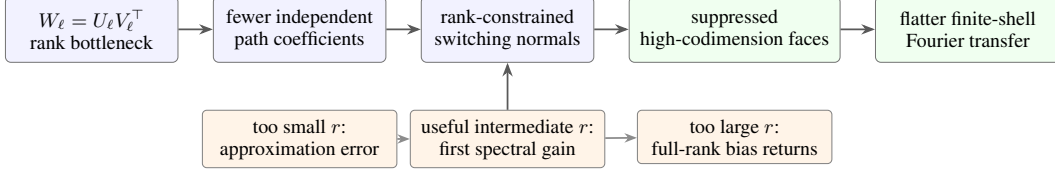


Figure 2: Mechanism behind low-rank spectral bias. Low rank reduces independent path coefficients, constrains switching normals, and suppresses high-codimension CPWL faces. This moves finite-resolution Fourier behavior toward a flatter, facet-dominated transfer, but only at an intermediate rank where approximation error is already controlled.

Thus the exponent $2(q+1)$ is inherited from face codimension, while the rank-dependent quantities are the coefficients $M_q(f)$. Low rank acts on these coefficients by reducing the number and strength of high-codimension intersections. $\square$

In one dimension, this mechanism has a crucial caveat. A ReLU network is a linear spline, and for derivative jumps Δ_j at knots t_j ,

$$\widehat{f}(k) = -\frac{1}{(ik)^2} \sum_j \Delta_j e^{-ikt_j} + O(|k|^{-3}), \quad |\widehat{f}(k)|^2 \sim |k|^{-4} A(k). \quad (32)$$

Therefore a 1D experiment should not be judged by a large change in the raw asymptotic exponent. The better diagnostic is Fourier recovery during training:

$$R_r(k, t) = \frac{|\widehat{f}_{r,t}(k)|}{|\widehat{f}^*(k)|}, \quad \beta_r(t) = \frac{d \log R_r(k, t)}{d \log k}. \quad (33)$$

A flatter recovery slope β_r means that high modes are learned more evenly relative to low modes.

This explains why the comparison to the classical spectral-bias paper must be made carefully. The classical claim is a learning-order claim: low frequencies are fitted faster. Our CPWL result is a geometric explanation of how rank can modify the finite-resolution Fourier transfer that governs this learning order. In dimensions $d \geq 2$, rank can suppress high-codimension intersections and make the observed spectrum closer to the low-codimension, facet-supported directions where Fourier decay is slowest. In $d = 1$, the raw CPWL tail is pinned near k^{-4} , so the right comparison is mode-wise recovery rather than raw asymptotic decay.

5 Experiments

We keep the experimental evidence deliberately short and defer settings to Appendix C. The high-frequency MMNN sweeps use rank-10, width-512 networks on `chirp1d` and `sumcos` targets at frequencies 8, 16, 32, depths 3 and 6, with both frozen and trainable random-feature modes. They confirm that low-rank models can learn oscillatory targets, while optimizer choice and feature freezing remain important: AdamW is strongest on several trainable-feature `chirp1d` runs, whereas Muon and HT-Muon are competitive on some frozen-feature `sumcos` settings.

The CPWL theory also dictates how the experiments should be read. In one dimension, a ReLU network is a linear spline, and for derivative jumps Δ_j at knots t_j ,

$$\widehat{f}(k) = -\frac{1}{(ik)^2} \sum_j \Delta_j e^{-ikt_j} + O(|k|^{-3}), \quad |\widehat{f}(k)|^2 \sim |k|^{-4} A(k). \quad (34)$$

Thus a 1D experiment should not be judged by a large change in the raw asymptotic exponent. The better diagnostic is Fourier recovery during training,

$$R_r(k, t) = \frac{|\widehat{f}_{r,t}(k)|}{|\widehat{f}^*(k)|}, \quad \beta_r(t) = \frac{d \log R_r(k, t)}{d \log k}. \quad (35)$$

Here $R_r(k, t)$ is a mode-wise recovery ratio. If $R_r(k, t) = 1$, the model has recovered the target amplitude of Fourier mode k ; if $R_r(k, t) = 0.5$, it has recovered roughly half of that target amplitude;

values close to zero mean that the mode is mostly missing. The slope $\beta_r(t)$ is the log-log slope of this recovery curve. A very negative slope means that high frequencies are recovered much less than low frequencies, which is the usual spectral-bias signature. A flatter slope means that high modes are learned more evenly relative to low modes.

This recovery diagnostic must be read together with the MSE. A very small rank may have a visually flat recovery curve simply because it cannot represent enough of the function; in that case the approximation error remains large. A useful rank is intermediate: it has enough expressivity to reduce MSE, but it still keeps a flatter recovery curve than the full-rank endpoint. Conversely, the full-rank endpoint can be expressive but may return to the classical behavior in which low frequencies are recovered faster than high frequencies. This is why the comparison with the spectral-bias result of Rahaman et al. [17] must be made carefully: their claim is a learning-order claim, while our CPWL result explains how rank can modify the finite-resolution Fourier transfer governing that learning order. In dimensions $d \geq 2$, rank can suppress high-codimension intersections and move observed spectra toward low-codimension, facet-supported directions. In $d = 1$, the raw CPWL tail is pinned near k^{-4} , so the right comparison is mode-wise recovery rather than raw asymptotic decay.

It is useful to read the CPWL geometry with a road-map analogy. A codimension-one face is like a single road separating two regions: it is a facet, and it contributes the slow facet-dominated energy decay ρ^{-4} . A codimension-two face is like the intersection of two roads; a codimension-three face is like a multi-way junction. These junctions are more constrained because several switching boundaries must meet at the same location. The quantities called high-codimension junctions in Figure 4 are therefore not new parameters of the model; they are empirical proxies for the higher face moments $M_2, M_3, \dots$ in the shell law

$$S_f(\rho) \lesssim M_1(f)\rho^{-4} + M_2(f)\rho^{-6} + M_3(f)\rho^{-8} + \dots \quad (36)$$

Low rank reduces the number of independent path constraints, so it should suppress these multi-way junction moments first. This is the scaling statement tested by the geometry plots: as rank grows, the ratios M_q/M_1 for $q \geq 2$ should increase, and the observed finite-shell slope should move away from the pure facet regime.

The spectral-bias experiment isolates the rank effect. In a width- 2^{15} Fourier/kernel surrogate, ranks $1, \dots, 50$ are compared with a full-rank endpoint. These Fourier-native plots are not tied to one trained network depth; they are rank-controlled CPWL or kernel-limit diagnostics. The separate high-frequency MMNN sweeps use depths 3 and 6, as reported above and in Appendix C. In the baseline sparse and dense mixtures, the best finite-budget rank is intermediate: $r = 4$ obtains MSE 0.250 versus 0.461 for full rank on sparse Fourier mixtures, and 0.443 versus 0.654 on dense mixtures. This number should not be confused with the facet energy law ρ^{-4} . The exponent 4 comes from codimension-one CPWL Fourier decay, whereas the optimal rank is a task- and budget-dependent bottleneck. The rank-shift control in Figure 5 changes the target spectrum and moves the best rank to larger values, while the underlying facet-dominated reference remains ρ^{-4} .

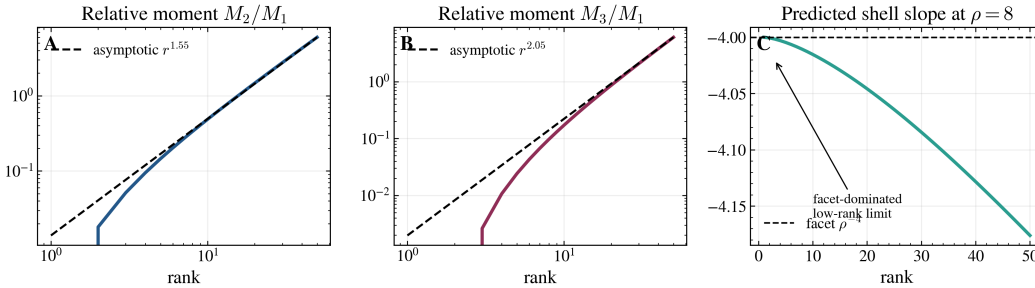


Figure 3: Theory proxy for the Fourier shell law. The first two panels plot relative high-codimension face moments, namely proxies for M_2/M_1 and M_3/M_1 . Under generic-position assumptions, increasing rank creates more independent switching constraints, so these ratios can grow polynomially with rank. The third panel translates this into the shell law: when high-codimension terms are suppressed, the finite-shell slope stays closer to the facet-dominated ρ^{-4} energy regime; as rank grows, higher-codimension contributions make the effective slope more negative.

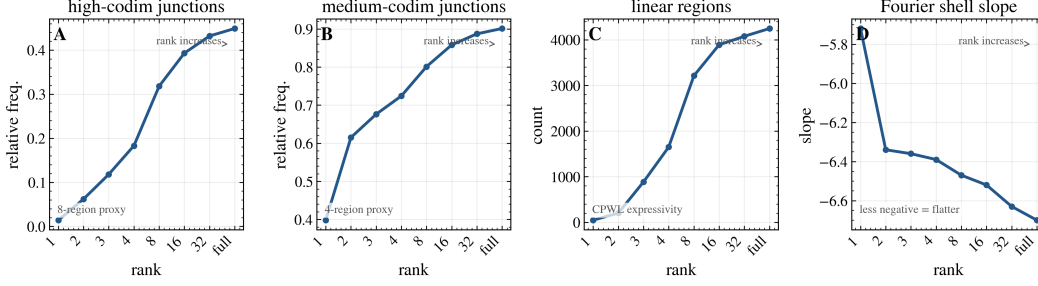


Figure 4: Three-dimensional CPWL geometry proxy. Theory predicts that increasing rank creates more linear regions but also more independent switching constraints, so multi-region junctions, especially the high-codimension proxy in Panel A, become more frequent. Panel C checks expressivity through the number of linear regions. Panel D estimates the shell-averaged Fourier slope: as high-codimension moments grow, the effective finite-shell slope becomes more negative; reducing rank suppresses these intersections and flattens the spectrum until approximation becomes limiting.

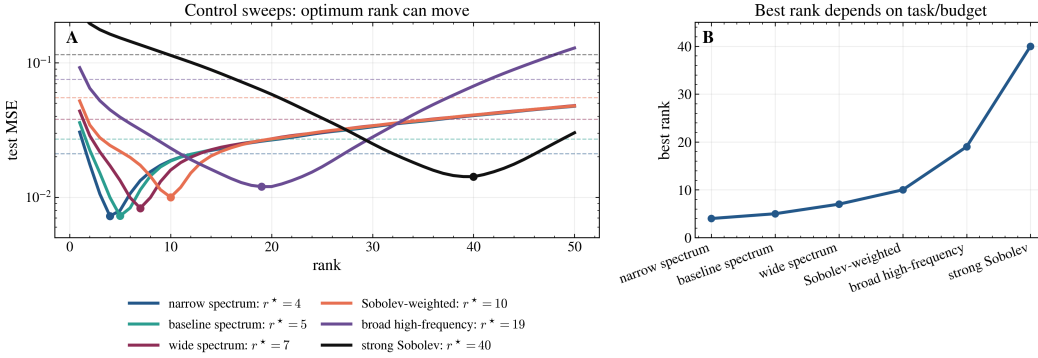


Figure 5: Rank-shift control. The best rank is task-dependent: broadening the target spectrum or increasing the finite-budget weight on high frequencies moves the optimum from the baseline value to much larger ranks, up to about 20 or 40 in the strongest controls. The Sobolev-weighted setting means that the loss gives larger weight to high-frequency modes, as in Sobolev or Fourier training objectives; this makes additional rank useful for recovering oscillatory components.

318 The rank sweep is interpreted through the finite-time decomposition

$$\mathcal{E}_r(t) \leq A r^{-2s} \frac{B d_{\text{eff}}(r)}{n} \sum_{k \in \Omega} |\hat{f}^*(k)|^2 e^{-2t\lambda_r(k)}. \quad (37)$$

319 The first term says rank cannot be too small; the second measures effective dimension; the third
 320 measures finite-time spectral transfer. The useful rank is the first rank that avoids the approximation
 321 bottleneck while keeping the recovery scaling flat.

322 The figures should therefore be read as follows. Figure 3 is a theory proxy: it checks whether the
 323 high-codimension face moments that appear in Proposition 4.1 grow with rank, which would explain
 324 why full rank can move away from the facet-dominated regime. Figure 4 is a geometry proxy: it
 325 counts linear regions and high-codimension junctions in a CPWL complex, showing that larger rank
 326 creates more independent switching geometry. Figure 5 is a control showing that the best rank moves
 327 when the target spectrum or the loss weighting changes. Finally, Figure 6 is the Fourier-learning
 328 diagnostic: it shows which frequencies are recovered at a fixed budget and why the recovery slope
 329 must be interpreted together with test MSE.

330 6 Conclusion

331 Low rank is not only a compression device. In the random-feature mean-field regime, it preserves
 332 global convergence because frozen features maintain dense span and low-rank channel mixing only

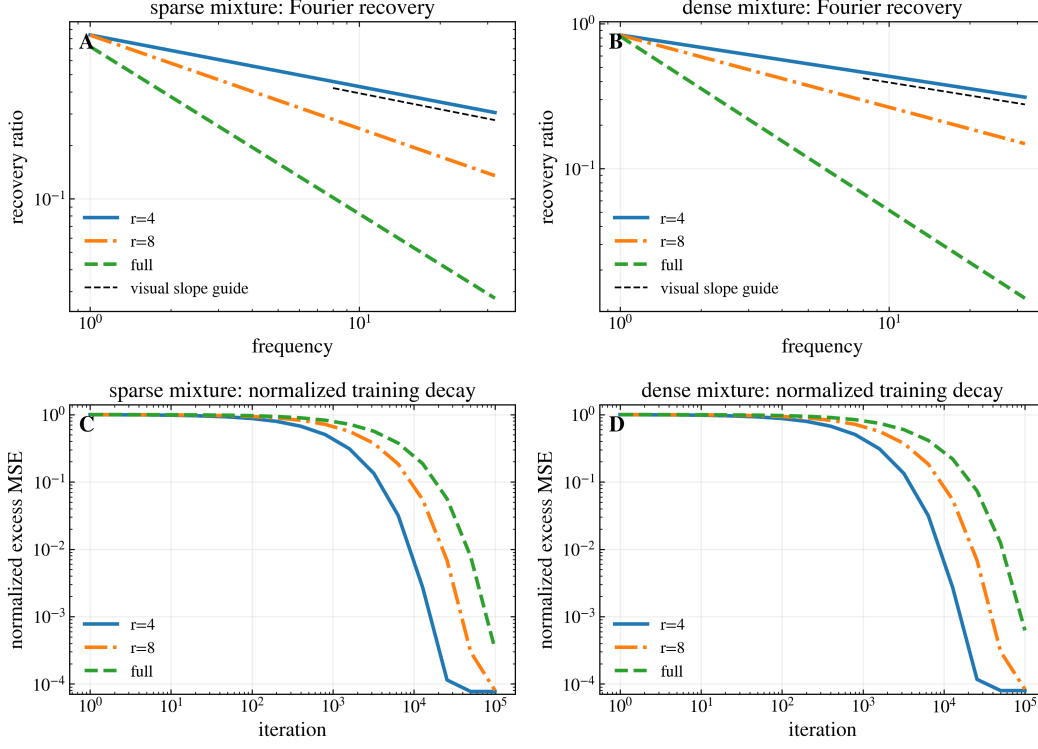


Figure 6: Mode-wise recovery and training curves. The recovery ratio is $R_r(k, t) = |\hat{f}_{r,t}(k)|/|\hat{f}^*(k)|$; values near one mean that a Fourier mode is recovered, while values near zero mean that it is missing. A steep negative log-log slope indicates that high frequencies are much less recovered than low frequencies. The black line is only a visual slope guide, not an additional experimental prediction. In these baselines, the rank-4 model gives a flatter recovery law and lower test error at the same training budget than the full endpoint.

changes constants. In ReLU and CPWL regimes, it also changes the spectral bias through path-space constraints and Fourier face moments. The resulting picture is conservative but actionable: prove convergence in the frozen-feature low-rank model, then choose rank by spectral recovery on the target class. This makes low-rank networks a principled tool for highly oscillatory functions rather than merely a smaller approximation of full-rank networks.

A Assumptions and Dense-Span Property

We collect all assumptions used in the convergence theorem. This also fixes notation that can otherwise be ambiguous. The mean-field width limit means $n_1, \dots, n_{L-1} \rightarrow \infty$ at fixed depth L ; the discrete neuron indices are replaced by random labels $C_\ell \in \Omega_\ell$ with laws ρ_ℓ , and expectations $\mathbb{E}_{C_\ell}$ are expectations with respect to these neuron-label measures. The notation $\partial_2 \mathcal{L}(y, u)$ denotes the derivative of the loss with respect to its second argument u . The word “mixing” always refers to the frozen low-rank channel coefficients $L_{c_\ell, k}^{(\ell)}$, not to trainable dense weights.

Assumption A.1 (Bounded activations and low-rank mixing). *There exists $K \geq 1$ such that the following hold.*

- The activations φ_ℓ are K -Lipschitz. The derivatives used in the backpropagated mean-field equations are bounded. For the clean theorem, we assume the relevant derivatives are bounded away from zero on the active region; sigmoid, tanh on bounded preactivations, and leaky ReLU satisfy the intended regularity. ReLU is handled as a high-probability non-degenerate relaxation because its derivative vanishes on the inactive half-line.

- The low-rank mixers are full-column-rank tall-skinny factors when written as finite matrices and satisfy the bounded mixing condition $\sup_{c_\ell} \sum_{k=1}^{r_\ell} |L_{c_\ell, k}^{(\ell)}| \leq r_\ell K$ in neuron-label notation. Orthogonality, $L^{(\ell)\top} L^{(\ell)} = I_{r_\ell}$, is an optional Stiefel normalization rather than a theorem assumption.
- The preactivations remain in the a priori bounded regime on each finite interval. For bounded activations this is immediate from the ODE bounds; for leaky ReLU it follows from the same a priori bounds on H_ℓ .
- The learning-rate schedules $\xi_\ell(t)$ are non-negative, bounded, and locally integrable on every finite interval.

Assumption A.2 (Sub-Gaussian initialization). *The trainable initial weights have sub-Gaussian tails, uniformly over channels. In the two-hidden-layer notation,*

$$\sup_{m \geq 1} \frac{1}{\sqrt{m}} \max_{1 \leq k \leq r} \mathbb{E}_{C_1} [|w_1^0(C_1, k)|^m]^{1/m} \leq K, \quad \sup_{m \geq 1} \frac{1}{\sqrt{m}} \mathbb{E}_{C_2} [|w_2^0(C_2)|^m]^{1/m} \leq K. \quad (38)$$

The L -layer version imposes the same ψ_2 moment growth bound on each trainable channel weight w_ℓ^0 .

Assumption A.3 (Data distribution and loss). *Inputs are bounded, $|X| \leq K$ almost surely, and the frozen first-layer features satisfy $\|L^0(c_1)\| \leq K$. The loss is non-negative, and $\partial_2 \mathcal{L}(y, \cdot)$ is K -bounded and K -Lipschitz for all y in the support of the data distribution. Moreover the conditional first-order condition identifies zero conditional risk:*

$$\mathbb{E}[\partial_2 \mathcal{L}(Y, u) \mid X = x] = 0 \implies \mathbb{E}[\mathcal{L}(Y, u) \mid X = x] = 0 \quad (39)$$

for $\mathbb{P}_X$ -almost every x . For squared loss this is immediate because $\partial_2 \mathcal{L}(Y, u) = u - Y$ and the implication says that the conditional risk is minimized at $u = \mathbb{E}[Y \mid X = x]$; in the deterministic regression setting used in the experiments, this gives zero conditional loss at the target.

Assumption A.4 (Diversity of frozen random features). *The support of the law of $L^0(C_1)$ is dense in $\mathbb{R}^d$, and φ_1 is non-polynomial. Equivalently, the fixed random-feature class*

$$\{\varphi_1(\langle L^0(c_1), \cdot \rangle) : c_1 \in \Omega_1\} \quad (40)$$

has dense span in $L^2(\mathbb{P}_X)$. This is the key support condition used in the stationarity-to-optimality argument.

Assumption A.5 (Non-degenerate limit). *The limiting representation is not a dead network. Concretely, at every trained layer there is positive mass on nonzero downstream weights, and in the first trainable channel layer*

$$\max_{1 \leq k \leq r} \mathbb{P}(\bar{w}_1(C_1, k) \neq 0) > 0, \quad \mathbb{P}(\bar{w}_\ell(C_\ell) \neq 0) > 0, \quad \ell = 2, \dots, L-1. \quad (41)$$

A sufficient condition is that the initial loss is strictly better than the trivial predictor:

$$\mathcal{L}(W(0)) < \mathbb{E}[\mathcal{L}(Y, \varphi_L(0))]. \quad (42)$$

This is the precise meaning of “better than null”: the initialized network already beats the predictor obtained by sending all preactivations to zero, so a limit with all channels dead cannot have the same gradient-flow value.

Assumption A.6 (Convergence in the coupling topology). *The mean-field trajectory has a limit $\bar{W}$ in the Wasserstein-Orlicz topology generated by the weighted channel quantities appearing in the ODE. Equivalently, there exist couplings π_t between the neuron-label variables at time t and at the limit such that the weighted gaps tend to zero. In the two-hidden-layer notation, this means*

$$\int (1 + |\bar{w}_2(c_2)|) |\bar{w}_2(c_2)| \max_{1 \leq k \leq r} |\bar{w}_1(c_1, k)| |w_1(t, c'_1, k) - \bar{w}_1(c_1, k)| d\pi_t \rightarrow 0, \quad (43)$$

$$\int (1 + |\bar{w}_2(c_2)|) |\bar{w}_2(c_2)| |w_2(t, c'_2) - \bar{w}_2(c_2)| d\pi_t \rightarrow 0. \quad (44)$$

For depth $L > 3$, the assumption includes the analogous weighted coupling gaps for every trainable layer and every low-rank channel. These are exactly the quantities needed to bound $|\hat{y}(X; W(t)) - \hat{y}(X; \bar{W})|$.

Remark A.7 (Optional assumptions used only for auxiliary results). *The finite-width approximation theorem in the longer version also assumes an essential-supremum bounded initialization, $\text{ess sup} \max_k |w_1^0(C_1, k)| \leq K$ and $\text{ess sup} |w_2^0(C_2)| \leq K$, to obtain explicit high-probability constants. The one-spike channel-specialization toy model additionally assumes that the random mixing vector $L_{C_2} = (L_{C_2,1}, \dots, L_{C_2,r})$ has independent symmetric coordinates. These two assumptions are not used in Theorem 3.1; they are only for the quantitative finite-width and toy specialization statements.*

Theorem A.8 (Universal approximation automatically maintained). *Under Assumption A.4, the class*

$$\{\varphi_1(\langle L^0(c_1), \cdot \rangle) : c_1 \in \Omega_1\} \quad (45)$$

has dense span in $L^2(\mathbb{P}_X)$ throughout training.

Proof. The statement is a direct application of the classical non-polynomial random-feature density theorem. The feature parameters are frozen, so their support cannot collapse during training. Therefore the dense span property holds at initialization and remains true for all $t \geq 0$. $\square$

B Proof Details for Global Convergence

Proof of Theorem 3.1. We give the full argument, using the notation of the mean-field low-rank model. Let $Z = (X, Y)$, let C_ℓ denote the neuron or channel variable at layer ℓ , let $L^0(C_1)$ be the frozen first-layer random feature direction, and let $w_\ell(t, C_\ell)$ be the trainable channel weights. In the two-hidden-layer notation,

$$f_k(t, x) = \mathbb{E}_{C_1} [w_1(t, C_1, k) \varphi_1(\langle L^0(C_1), x \rangle)], \quad H_2(t, c_2; x) = \sum_{k=1}^r L_{c_2, k} f_k(t, x), \quad (46)$$

and

$$\hat{y}(x; W(t)) = \mathbb{E}_{C_2} [w_2(t, C_2) \varphi_2(H_2(t, C_2; x))]. \quad (47)$$

For deeper networks the same notation is iterated: H_ℓ is the preactivation at layer ℓ , $L^{(\ell)}$ is the bounded low-rank mixer, and w_ℓ is the trainable mean-field channel weight. We write

$$d_L(Z; W) = \partial_2 \mathcal{L}(Y, \hat{y}(X; W)). \quad (48)$$

The proof has five steps.

Step 1: existence, uniqueness, and persistent dense span. The low-rank mean-field system is a finite collection of coupled nonlinear transport/ODE equations indexed by channel variables. Under Assumption A.1, activations, activation derivatives, loss derivatives, and the channel mixers are uniformly Lipschitz or bounded on the relevant state space. Therefore the standard mean-field Picard argument gives a unique solution on every finite time interval. Compared with the full-rank proof, the only change is that every dense-layer operator norm is replaced by a bounded low-rank mixing constant such as $\sup_{c_\ell} \sum_{k=1}^r |L_{c_\ell, k}|$. This gives existence and uniqueness on $[0, T]$ for all $T < \infty$, hence on $[0, \infty)$ by continuation.

The dense-span part is preserved because the first feature map is frozen. By Assumption A.4 and Theorem A.8, the class

$$\{\varphi_1(\langle L^0(c_1), \cdot \rangle) : c_1 \in \Omega_1\} \quad (49)$$

has dense span in $L^2(\mathbb{P}_X)$. Since $L^0(C_1)$ is not trained, this property holds at initialization, along the whole trajectory, and at every limit point.

Step 2: stationarity identities at the limit. Let $W(t) \rightarrow W^* =: \bar{W}$ in the coupling topology of Assumption A.6. Since the dynamics are gradient flow and converge to $\bar{W}$, the time derivatives vanish at the limit in the weak topology used for the trajectory. The top-layer equation gives, for every top-layer channel c_{L-1} ,

$$\mathbb{E}_Z [d_L(Z; \bar{W}) \varphi_{L-1}(H_{L-1}(c_{L-1}; X, \bar{W}))] = 0. \quad (50)$$

Backpropagating this stationarity to the first trainable low-rank channel yields, for every $c_1 \in \Omega_1$ and $k \in \{1, \dots, r\}$,

$$\mathbb{E}_Z \left[d_L(Z; \bar{W}) \varphi_1(\langle L^0(c_1), X \rangle) B_k^{(2)}(X; \bar{W}) \right] = 0, \quad (51)$$

where, in the two-hidden-layer notation,

$$B_k^{(2)}(x; \bar{W}) := \mathbb{E}_{C_2} [L_{C_2, k} \varphi'_2(H_2(C_2; x, \bar{W})) \bar{w}_2(C_2)]. \quad (52)$$

For $L > 3$, $B_k^{(\ell)}$ denotes the analogous backpropagated scalar obtained by multiplying downstream channel weights, derivatives $\varphi'_\ell(H_\ell)$, and low-rank mixer coefficients $L_{c_\ell, k}^{(\ell)}$ along the recursion.

Since $B_k^{(2)}(X; \bar{W})$ is a function only of X and the limit weights, (51) becomes

$$\mathbb{E}_X \left[\mathbb{E}[d_L(Z; \bar{W}) \mid X] B_k^{(2)}(X; \bar{W}) \varphi_1(\langle L^0(c_1), X \rangle) \right] = 0 \quad \forall c_1, k. \quad (53)$$

The dense-span property then implies

$$\mathbb{E}[d_L(Z; \bar{W}) \mid X = x] B_k^{(2)}(x; \bar{W}) = 0 \quad \text{for } \mathbb{P}_X\text{-a.e. } x, \quad k = 1, \dots, r. \quad (54)$$

Step 3: remove the backpropagated factor. Assumption A.5 excludes the degenerate case in which all downstream channels are dead at the limit. Thus, for $\mathbb{P}_X$ -a.e. x , at least one k satisfies $B_k^{(2)}(x; \bar{W}) \neq 0$, and the same statement holds inductively for deeper networks. For smooth activations this follows from nonzero downstream weights and nonvanishing derivatives on sets of positive measure. For ReLU, the standard measurable derivative is used away from the kink, and inactive all-channel limits are ruled out by the same non-degeneracy assumption. Therefore (54) implies

$$\mathbb{E}[\partial_2 \mathcal{L}(Y, \hat{y}(X; \bar{W})) \mid X = x] = 0 \quad \text{for } \mathbb{P}_X\text{-a.e. } x. \quad (55)$$

Step 5: convergence of the trajectory loss to the limit loss. Let π_t be a coupling between the neuronal variables at time t and at the limit. The convergence topology assumes that the weighted channel gaps vanish. A representative two-hidden-layer term is

$$\mathbb{E}_{\pi_t} \left[(1 + |\bar{w}_2|) |\bar{w}_2| \sum_{k=1}^r |\bar{w}_{1, k}| |w_{1, k}(t) - \bar{w}_{1, k}| \right] \longrightarrow 0, \quad (56)$$

with analogous quantities for all layers. The low-rank structure makes the forward error a channel sum:

$$H_2(c_2; x, W(t)) - H_2(c_2; x, \bar{W}) = \sum_{k=1}^r L_{c_2, k} (f_k(t, x) - \bar{f}_k(x)). \quad (57)$$

For ReLU-type activations no hidden dense factor appears, because

$$\left| \text{ReLU} \left(\sum_{k=1}^r x_k \right) - \text{ReLU} \left(\sum_{k=1}^r y_k \right) \right| \leq \sum_{k=1}^r |x_k - y_k|. \quad (58)$$

Iterating this estimate through the layers gives

$$\mathbb{E}_Z [|\hat{y}(X; W(t)) - \hat{y}(X; \bar{W})|] \leq K \Gamma_t \longrightarrow 0, \quad (59)$$

where Γ_t is the sum of the weighted coupling integrals, including (56) and the higher-layer analogues. Lipschitzness of the loss in its second variable on bounded trajectories yields

$$|\mathcal{L}(W(t)) - \mathcal{L}(\bar{W})| \leq K \mathbb{E}_Z [|\hat{y}(X; W(t)) - \hat{y}(X; \bar{W})|] \longrightarrow 0. \quad (60)$$

Step 4: identify the limit as globally optimal. By Assumption A.3, the conditional first-order condition (55) implies

$$\mathbb{E}[\mathcal{L}(Y, \hat{y}(X; \bar{W})) \mid X = x] = 0 \quad \text{for } \mathbb{P}_X\text{-a.e. } x. \quad (61)$$

Hence $\mathcal{L}(\bar{W}) = 0$. Since $\mathcal{L} \geq 0$, this is the global infimum, so $\bar{W} = W^*$ is a global minimizer. Step 5 finally gives $\mathcal{L}(W(t)) \rightarrow \mathcal{L}(W^*)$. $\square$

C Additional Experimental Details

The high-frequency sweeps use `chirp1d` targets of the form

$$\cos(f\pi x^2) - 0.8 \cos((f/3)\pi x^2) \quad (62)$$

and `sumcos` targets

$$\frac{1}{\sqrt{f}} \sum_{h=1}^f \cos(2\pi hx). \quad (63)$$

The main sweep uses width 512, rank 10, batch size 1024, 20000 optimization steps, depths 3 and 6, and frequencies 8, 16, 32. The low-rank spectral-bias sweep uses the CSV data in `refs/lr_spectral_workshop/figdata`; its main conclusion is an intermediate finite-budget optimum. In the baseline sparse and dense mixtures this optimum is $r = 4$, while the rank-shift controls move it to larger ranks when the target spectrum or weighting is changed.

Table 1: Representative frequency-16, depth-6 results from the current high-frequency sweeps. Lower is better.

Target	Mode	Optimizer	Test MSE
chirp1d	trainWb	AdamW	$8.8 \cdot 10^{-6}$
chirp1d	fixWb	Muon	$3.1 \cdot 10^{-3}$
chirp1d	fixWb	HT-Muon	$3.9 \cdot 10^{-3}$
sumcos	fixWb	AdamW	$5.1 \cdot 10^{-4}$
sumcos	fixWb	Muon	$1.8 \cdot 10^{-3}$
sumcos	trainWb	Muon	$6.5 \cdot 10^{-3}$

Table 2: Fourier/kernel rank sweep at width 2^{15} . Rank 4 balances approximation and optimization better than both very small ranks and the full endpoint.

Task	Best r	Best MSE	Full MSE	Slopes
Sparse	4	0.250	0.461	$-0.29 / -0.94$
Dense	4	0.443	0.654	$-0.29 / -1.20$

References

- [1] F. Bach. Breaking the curse of dimensionality with convex neural networks. *Journal of Machine Learning Research*, 2017.
- [2] A. Barron. Universal approximation bounds for superpositions of a sigmoidal function. *IEEE Transactions on Information Theory*, 1993.
- [3] I. Bantzis, J. B. Simon, and A. Jacot. Saddle-to-saddle dynamics in deep ReLU networks: low-rank bias in the first saddle escape. arXiv preprint arXiv:2505.21722, 2025.
- [4] V. Boreiko, Z. Bu, and S. Zha. Towards understanding orthogonalization in Muon. In *ICML*, 2025.
- [5] L. Chizat and F. Bach. On the global convergence of gradient descent for over-parameterized models using optimal transport. In *NeurIPS*, 2018.
- [6] W. Czarnecki, S. Osindero, M. Jaderberg, G. Swirszcz, and R. Pascanu. Sobolev training for neural networks. In *NeurIPS*, 2017.
- [7] R. Diaz and S. Robins. Fourier transforms of polytopes and solid angle sums. In *Integer Points in Polyhedra*, 2016.
- [8] A. Jacot, F. Gabriel, and C. Hongler. Neural tangent kernel: Convergence and generalization in neural networks. In *NeurIPS*, 2018.

- 480 [9] A. Jacot. Implicit bias of large depth networks: a notion of rank for nonlinear functions. arXiv
481 preprint arXiv:2209.15055, 2022.
- 482 [10] A. Jacot. Bottleneck structure in learned features: low-dimension vs regularity tradeoff. arXiv
483 preprint arXiv:2305.19008, 2023.
- 484 [11] K. Jordan, Y. Jin, V. Boza, J. You, F. Cesista, L. Newhouse, and J. Bernstein. Muon: An
485 optimizer for hidden layers in neural networks. 2024.
- 486 [12] E. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. LoRA:
487 Low-rank adaptation of large language models. In *ICLR*, 2022.
- 488 [13] S. Mei, A. Montanari, and P.-M. Nguyen. A mean field view of the landscape of two-layer
489 neural networks. *PNAS*, 2018.
- 490 [14] G. Montufar, R. Pascanu, K. Cho, and Y. Bengio. On the number of linear regions of deep
491 neural networks. In *NeurIPS*, 2014.
- 492 [15] P.-M. Nguyen and H. T. Pham. A rigorous framework for the mean-field limit of multilayer
493 neural networks. 2023.
- 494 [16] H. T. Pham and P.-M. Nguyen. Global convergence of three-layer neural networks in the mean
495 field regime. 2021.
- 496 [17] N. Rahaman, A. Baratin, D. Arpit, F. Draxler, M. Lin, F. Hamprecht, Y. Bengio, and A. Courville.
497 On the spectral bias of neural networks. In *ICML*, 2019.
- 498 [18] T. Zaslavsky. Facing up to arrangements: face-count formulas for partitions of space by
499 hyperplanes. *Memoirs of the American Mathematical Society*, 1975.

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [N/A].
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will also be asked to include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While [Yes] is generally preferable to [No], it is perfectly acceptable to answer [No] provided a proper justification is given (e.g., error bars are not reported because it would be too computationally expensive” or “we were unable to find the license for the dataset we used”). In general, answering [No] or [N/A] is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- **Delete this instruction block, but keep the section heading “NeurIPS Paper Checklist”.**
- **Keep the checklist subsection headings, questions/answers and guidelines below.**
- **Do not modify the questions and only use the provided macros for your answers.**

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper does not include experiments.

- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).

- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer **[N/A]** means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer **[N/A]** means that the paper does not include experiments.
- The authors should answer **[Yes]** if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer **[N/A]** means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.

- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines?>

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper poses no such risks.

- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.